H.T.No: | | | | | | | | | | | Course Code: **241MA002**

# ADITYA UNIVERSITY

B.Tech – II Semester End Examinations Regular – Jun 2025

**DIFFERENTIAL EQUATIONS & VECTOR CALCULUS**
**(Common to all Except Ag.E)**

**Time: 3 hours** **Max. Marks: 100**

**Answer ONE question from each unit**
**All Questions Carry Equal Marks**
**All parts of the questions must be answered at one place only**

**UNIT-I**

| | | | | | |
|---|---|---|---|---|---|
| 1 | a | Solve the differential equation $(x^2+y^2)dx-2xydy=0$ | L2 | CO1 | [10M] |
| | b | Solve the differential equation $(1+x^2)\frac{dy}{dx}+y=e^{tan^{-1}x}$ | L2 | CO1 | [10M] |
| | | **(OR)** | | | |
| **2** | a | Find orthogonal trajectories of family of circles $x^2+y^2=2x$. | L2 | CO1 | [10M] |
| | b | A battery ofemf 60 volts is connected in series with a resistor of 6 ohms and an inductor of 3 Henries. If the switch is closed at time t=0, find the current I(t) in the circuit for t>0. | L2 | CO1 | [10M] |

**UNIT-II**

| | | | | | |
|---|---|---|---|---|---|
| 3 | a | Solve the differential equation $(D^2-5D+6)y=e^{2x}+\cos 3x$ | L2 | CO2 | [10M] |
| | b | Using method of variation of parameters, find the solution of the differential equation $\frac{d^2y}{dx^2}+y=\frac{cosx}{x}$. | L3 | CO2 | [10M] |
| | | **(OR)** | | | |
| 4 | a | Solve $x^2\frac{d^2y}{dx^2}+3x\frac{dy}{dx}+y=sinx$ | L2 | CO2 | [10M] |
| | b | Describe the formation of a second-order differential equation for a series LCR circuit. | L3 | CO2 | [10M] |

**UNIT-III**

| | | | | | |
|---|---|---|---|---|---|
| 5 | a | Solve the partial differential equation: $x(y-z)p+y(z-x)q=z(x-y)$ | L3 | CO3 | [10M] |
| | b | Solve the partial differential equation: $\frac{\partial^4 z}{\partial x^4}+\frac{\partial^4 z}{\partial y^4}=0$ | L3 | CO3 | [10M] |
| | | **(OR)** | | | |
| 6 | a | Solve the partial differential equation: $\frac{\partial^3 z}{\partial x^3}-2\frac{\partial^3 z}{\partial x\,\partial y^2}+3\frac{\partial^3 z}{\partial y^3}=sinx+y^2$ | L3 | CO3 | [10M] |
| | b | Solve the partial differential equation: $z(x+y)=px^2+qy^2$ | L3 | CO3 | [10M] |

**UNIT-IV**

| | | | | | |
|---|---|---|---|---|---|
| 7 | a | If $u=x-y+z, v=x^2+y^2-z^2, w=xz-yz+xy$, then find $[\nabla u\ \nabla v\ \nabla w]$ | L2 | CO4 | [10M] |
| | b | Find the directional derivative of $f(x,y,z)=x^2y+3yz^2$ at point $A(1,-1,2)$ in the direction of the line from A to B=(3,1,4) | L3 | CO4 | [10M] |

**(P.T.O)**

**(OR)**

| | | | | | |
|---|---|---|---|---|---|
| 8 | a | Find the divergence of the vector field $\vec{F}=\nabla(x^2y+y^2z+z^2x)$ | L2 | CO4 | [10M] |
| | b | Show that the vector field $\vec{F}=(2xz+y)i+(x+z^2)j+(2yz+x^2)k$ is irrotational, and hence find its scalar potential. | L3 | CO4 | [10M] |

**UNIT-V**

| | | | | | |
|---|---|---|---|---|---|
| 9 | a | Find the work done in moving a particle in the force field $\vec{F}=2xyi + xzj +yzk$ along the straight line from (1,0,0) to (2,1,3). | L2 | CO5 | [10M] |
| | b | Evaluate $\iint_S \vec{F}.\hat{n}\,dS$ where $\vec{F}=xi +yj +zk$ and S is the portion of the plane $x+y+z=6$ in the first octant. | L3 | CO5 | [10M] |

**(OR)**

| | | | | | |
|---|---|---|---|---|---|
| 10 | a | Verify Green's Theorem for $\oint_C (x^2y+y^2)\,dx+(2xy+x)\,dy$ where C is the closed curve bounding the region enclosed between the parabola $y=x^2$ and the line $y=4$. | L2 | CO5 | [15M] |
| | b | State i) Gauss divergence theorem ii) Stoke's theorem | L3 | CO5 | [5M] |

*****